Current State of the EA and Barcoding Detector

What happens as an OCT scan moves through the pipeline?

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One Scan, One Traceable Decision Path

Load the selected B-scan $\longrightarrow$ Standardize the image $\longrightarrow$ Measure visible patterns

$\longrightarrow$ Score EA and barcoding separately $\longrightarrow$ Reject likely structure

$\longrightarrow$ Report intervals and confidence evidence

The output assigns each horizontal location to **normal**, **EA**, or **barcoding**. Adjacent positive columns become an interval whose location and width can be reported. Every rejection stage records why a candidate was removed.

Step 1: Loading the Relevant Scan

Input

The validation workflow opens the native HEYEX E2E volume and selects the patient-specific B-scan number recorded in the manual annotation.

Anatomical reference

The scan is loaded together with its Bruch membrane (BM) boundary. This keeps the image and the scanner-provided retinal landmark paired throughout processing.

The same loader can accept a PNG with JSON metadata, but the current validation results begin with E2E volumes so neighboring B-scans and BM metadata remain available.

Step 2: Standardizing the B-scan

The selected preprocessing configuration is fixed across all ten cases:

Operation	**Selected setting**	**Result**
Alignment	Flatten to BM	BM appears at a common depth
Region retained	150 pixels below BM	Same anatomical depth range
Brightness scaling	Within-scan z-score	Relative brightness values
Denoising	Gaussian $(1.0, 0.5)$	Mild noise reduction; stripes preserved
Incomplete depth	Padding allowed	Scan remains processable

The detector receives a consistently aligned, approximately 150×512-pixel sub-BM image.

Step 3: Turning Appearance into Measurements

At every horizontal position, the detector asks a small set of clinically readable questions:

- Is the tissue column broadly brighter than the rest of this scan?
- How bright are its brightest relevant pixels?
- Does the pattern remain aligned as it extends deeper?
- Does it form an organized vertical structure?
- Does it contain repeated bright–dark stripe texture?
- Is the brightness concentrated near BM, in the middle, or deeper?

Each measurement is expressed relative to the same scan. This allows the final JSON to show which visible properties supported each interval.

The Detector Architecture

1. **Candidate scores:** independent EA and barcoding models score every column from brightness, continuity, vertical organization, and their combinations.
2. **Local context:** isolated responses are suppressed unless nearby columns provide compatible evidence.
3. **Interval formation:** short gaps may be joined and very short detections are removed.
4. **Structural veto:** likely vessel/structural shadows and dark continuous columns are excluded.
5. **Barcoding confirmation:** the complete interval must pass stripe-texture and near-depth brightness checks.
6. **Output:** remaining intervals are labeled EA or barcoding; all other columns are normal.

The stages are ordered so that a structural region cannot be filled back into a disease interval after it has been rejected.

Different Criteria for EA and Barcoding

barcoding

- Probability threshold: 0.55
- Minimum interval: 8 columns
- Local support radius: 5 columns
- Requires 2 of 3 feature votes
- Gabor mean/peak: 0.40/0.50
- Near-depth mean: at least 0.0

EA

- Probability threshold: 0.50
- Minimum interval: 12 columns
- Local support radius: 7 columns
- Requires all 3 feature votes
- No Gabor requirement
- Provisional depth gate disabled

These are phenotype-specific decisions: the best barcoding settings are not automatically applied to EA.

How Likely Vessel or Structural Signal Is Rejected

Two complementary checks are applied after EA/barcoding candidates form:

- A structural model combines intensity, vertical organization, and the amount of structural edge signal. Its veto threshold is 0.90.
- A direct dark-shadow rule identifies vertically continuous columns whose upper brightness is unusually low—a pattern more consistent with shadowing than hypertransmission.
- Each rejected run is expanded slightly to include its immediate shadow boundary.

Remaining challenge

Some reproducible normal anatomy still resembles disease. It may persist into neighboring scans and can also have strong texture and depth measurements.

What the Gabor and Depth Checks Added

Stripe-texture evidence

Three pattern widths—4, 8, and 16 pixels—are examined. A candidate must show repeated vertical bright–dark variation across the interval, rather than a single bright column.

Depth evidence

The sub-BM image is summarized in near (0–20%), middle (20–50%), and deep (50–100%) bands. The selected barcoding rule requires at least average near-band brightness.

Together these checks preserved all 424 V3 true-positive barcoding columns while reducing false-positive columns from 302 to 95.

Adjacent B-scans: Confidence, Not a Mandatory Veto

For each interval, the detector checks two neighboring B-scans on either side for a same-label interval at a similar horizontal location.

Observed support	**Suggested interpretation**
Both sides	Higher volume-consistency confidence
At least one neighbor	Moderate volume-consistency confidence
No neighboring match	Lower confidence; retain for now

Mandatory adjacency improved precision but removed true lesions. Persistent vessel/anatomical patterns also survived it. Adjacent support is therefore saved as explanatory evidence rather than used in the selected final gate.

Validation Set and How to Read the Scores

Ten selected B-scans were manually annotated: five fast-progressing and five slow-progressing subjects. Uncertain and vessel/structural areas are excluded from scoring.

Precision

Of the columns marked by the detector, how many were annotated as that phenotype?

Recall

Of the manually marked phenotype, how much did the detector recover?

Dice

A combined measure of overlap, false alarms, and misses. One is perfect.

Important: these ten scans also informed parameter selection. The results describe current development performance, not clinical validation.

Current Best Aggregate Results

Class	**Configuration**	**Precision**	**Recall**	**Dice**
Barcoding	V2	0.526	0.506	0.516
Barcoding	Contextual V3	0.584	0.520	0.550
Barcoding	Initial Gabor	0.777	0.458	0.576
Barcoding	**Gabor + depth**	**0.817**	**0.520**	**0.636**
EA	V2	0.369	**0.739**	0.492
EA	**Contextual V3**	**0.475**	0.699	**0.566**
EA	Gabor + barcoding depth	0.462	0.699	0.557

The best configuration is phenotype-specific: Gabor + depth for barcoding, and contextual V3 without the experimental EA depth gate for EA.

What Worked—and What Did Not

Useful additions

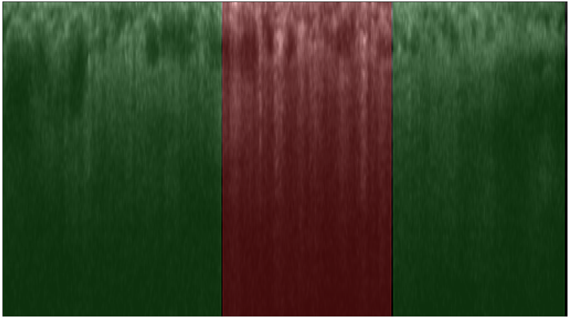
- Separate EA/barcoding calibration
- Local spatial support
- Structural and dark-shadow vetoes
- Multi-scale stripe texture
- Near-depth barcoding evidence

Not selected

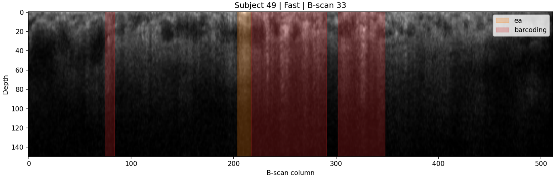
- Additional learned normal model
- Mandatory adjacent support
- Bilateral volume veto
- Conservative hybrid rejection
- Balanced hybrid rejection

The hybrid rules combined absent neighbor support, short width, and weak texture/depth/probability. They removed true intervals faster than they removed false ones; the control retained the best Dice.

Strong Barcoding Case: Fast 49, B-scan 33



Manual annotation

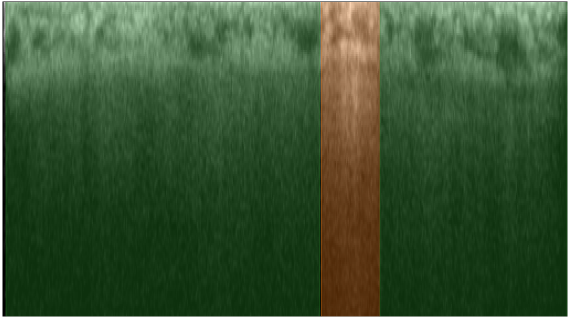


Automatic output

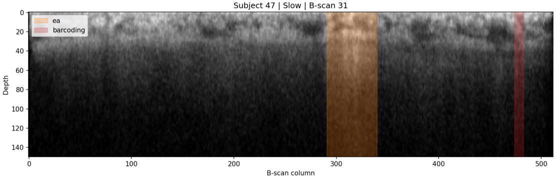
barcoding: precision 0.93, recall 0.78, Dice 0.85.

The detector localizes the major barcoding regions closely, with relatively little additional marked tissue.

Strong EA Case: Slow 47, B-scan 31



Manual annotation

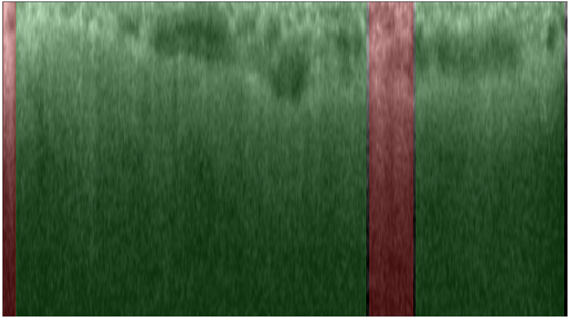


Automatic output

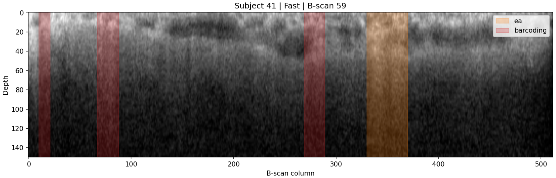
EA: precision 1.00, recall 0.91, Dice 0.95.

The automatic interval nearly reproduces the location and extent of the manually marked EA region.

Weak Barcoding Case: Fast 41, B-scan 59



Manual annotation

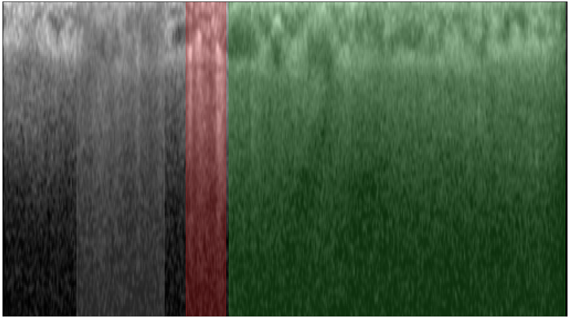


Automatic output

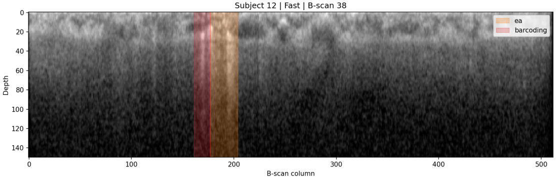
barcoding: precision 0.06, recall 0.06, Dice 0.06.

Repeatable structural patterns survive the current veto and account for 49 of the 95 remaining false-positive barcoding columns across the development set.

Another Weak Case: Fast 12, B-scan 38



Manual annotation



Automatic output

barcodingDice 0.44; a false EA interval is also marked.

This case illustrates both incomplete recovery of barcoding and confusion between a bright structural region and EA.

Why the Hybrid and Volume Rules Were Not Selected

Barcoding rule	**Precision**	**Recall**	**Dice**
Single-scan control	0.817	**0.520**	**0.636**
Any one neighbor	0.872	0.479	0.618
Bilateral neighbors	**0.982**	0.336	0.501
Hybrid conservative	0.810	0.497	0.616
Hybrid balanced	0.835	0.479	0.608

The conservative hybrid removed 19 true columns and no false columns. The balanced version removed 18 false columns but also 34 true columns. Current texture, depth, probability, and adjacency evidence cannot reliably distinguish every structural false positive from a true short lesion.

Current Output for Each Scan

For every processed B-scan, the pipeline now produces:

- a PNG overlay showing EA and barcoding intervals;
- the number of intervals for each phenotype;
- each interval's start, end, and width in pixels;
- EA and barcoding probabilities and contextual decisions;
- texture and near/middle/deep brightness evidence;
- structural and dark-shadow exclusions;
- matching intervals in adjacent scans and a volume-support count.

This provides a trace from the final highlighted interval back to the evidence that allowed it to remain.

Future Direction

1. **Add genuinely different structural evidence.** Compare the vertical brightness profile inside a candidate with neighboring normal tissue. Vessel shadowing should darken deeper signal, whereas hypertransmission should brighten it.
2. **Keep volume support as confidence.** Report bilateral, one-neighbor, or unsupported status without automatically deleting a lesion.
3. **Expand independent annotation.** Additional patients and clinician review are more valuable now than further tuning on the same ten scans.
4. **Lock and test parameters.** Evaluate the selected settings on untouched subjects before interpreting them as reliable biomarkers.

Current selected state

Contextual V3 for EA; contextual V3 plus Gabor and near-depth evidence for barcoding. Adjacent evidence is explanatory, not a hard rejection rule.

Take-Home Message

- The scan moves through a transparent sequence of anatomical alignment, visible feature measurement, phenotype-specific scoring, and structural rejection.
- Gabor texture plus near-depth evidence produced the largest barcoding improvement: precision 0.817 and Dice 0.636.
- EA currently performs best with contextual V3: precision 0.475, recall 0.699, and Dice 0.566.
- More aggressive volume and hybrid rejection increased selectivity but removed too much real disease signal.
- The main limitation is distinguishing persistent normal/vascular anatomy from true pathology with only ten non-clinician-annotated scans.

The next improvement should add new anatomical evidence, not merely recombine the current scores.